single, interactive entity. Borrowing a concept from engineering, he saw the network of interactions as a dynamic, physical system. On the suggestion of his colleague Arthur Clapham, he coined the word “ecosystem” to describe it.

This idea had been developing long before Arthur Tansley published his influential paper on the subject in 1935. As early as 1864, conservationist George Perkins Marsh, in his book *Man and Nature*, had identified “the woods,” “the waters,” and “the sands” as different types of habitat. He examined how the relationship between them and the animals and plants that lived in them could be upset by human activity.

Interconnected systems

By the 20th century, the idea had taken hold that these and other environments could be understood as discrete entities, with distinctive interactions between the living and inert elements within them (although there are also important interactions between different ecosystems). In 1916, American ecologist Frederic Clements built on this idea in his work on plant succession, referring to a “community” of vegetation as a single unit, and using the term “biome” to describe the whole complex of organisms inhabiting a given region.

Tansley envisaged ecosystems as being made up of biotic (living) elements and abiotic (nonliving) elements such as energy, water, nitrogen, and soil minerals, which are essential to the functioning of the systems as a whole. The biotic components within an ecosystem not only interact with one another, but also with the abiotic parts. Thus, within any given ecosystem, the organisms adapt to both the biological and physical elements of the environment. The different types of ecosystem can be defined by their physical environments. There are four categories of ecosystem: terrestrial, freshwater, marine, and atmospheric. These can be further subdivided into various types according to different physical environments